COMMUNICATIONS AND INFORMATION SERVICE

StationNet

Official User's Guide and Operator's Reference

For StationNet Version 1.0

First Edition

November 2026

Notice

This manual describes StationNet Version 1.0 as distributed in November 2026. StationNet is under continuing development, and later releases may differ in detail from the descriptions given here.

StationNet is modular. Each station is set up by its own operator, who decides which services are offered, who may use them, and what material they contain. The services, channels, books, games, programs, and worlds you find on a particular station may therefore differ from the examples in this manual.

Screen names, button captions, and commands are printed exactly as they appear on the screen. Where the station operator may change a setting, the manual says so.

Product names mentioned in this manual are used for identification only and may be trademarks of their respective owners.

Preface

StationNet is a self-hosted communications and information service. A single computer on a home, school, club, or office network runs the station, and everyone else reaches it with an ordinary web browser. No outside service, subscription, or Internet connection is required once a station is installed.

This manual is written first for the people who use a station: those who read the bulletins, exchange mail, listen to the radio, read in the library, work in the Office programs, or play in the MUD. Parts I through V assume no knowledge of StationNet beyond basic familiarity with a computer and a web browser.

Parts VI and VII serve two smaller audiences. Part VI is for the station operator, called the SYSOP, who installs StationNet, starts and stops stations, and administers accounts and content. Part VII is a short orientation for developers who wish to write programs, themes, or game worlds for StationNet.

How This Manual Is Organized

Part I, Getting Started, explains what a station is, how to connect, how to create an account, and how the screen is arranged. It ends with a short guided session.

Part II, Communication and Community, covers the user directory, chat, private mail, the bulletin board, the imageboard, events, and the file archive.

Part III, Information and Entertainment, covers radio, television, videos, the library, the reference library (WIKI), programs, the arcade, and the MUD.

Part IV, Office, covers the six Office programs: Documents, Sheets, Presentations, Draw, Agenda, and BASIC.

Part V, Personalization and Everyday Use, collects the settings and habits that apply to every service: appearance, sound, notifications, installation on a phone or desktop, station hours, and account security.

Part VI, Station Hosting and Administration, is for SYSOPs.

Part VII, Developer's Reference, is for programmers.

The **Appendices** contain a keyboard reference, a quick reference to common operations, a troubleshooting guide, a table of modules and permissions, and a glossary.

Conventions Used in This Manual

`MONOSPACED CAPITALS` show text that appears on the screen, such as a tab name or button caption: for example, choose `COMPOSE +`.

`Monospaced lower case` shows something you type exactly, such as a MUD command (`/look`) or a file name (`station.json`).

Words in angle brackets stand for a value you supply. In `/go <direction>`, replace `<direction>` with an actual direction, such as `north`.

Choose means click with the mouse or tap on a touch screen. **Press** refers to a key on the keyboard.

Three kinds of notices appear in the text:

NOTE	A NOTE gives additional information or a helpful suggestion.

CAUTION	A CAUTION warns of an action that may cause inconvenience or loss of work.

WARNING	A WARNING identifies an action that permanently destroys information or affects other people's access to the station.

Contents

PART I. GETTING STARTED 7
1. Introducing StationNet 7
2. Connecting and Signing In 8
3. The Station Workspace 9
4. Your Profile and Account 11
5. A First Session 12

PART II. COMMUNICATION AND COMMUNITY 13
6. Users and Profiles 13
7. Chat 13
8. Private Mail 14
9. Bulletin Board, Announcements, and Polls 15
10. Imageboard 15
11. Events 16
12. File Archive 17

PART III. INFORMATION AND ENTERTAINMENT 18
13. Radio 18
14. Television 19
15. Videos 20
16. Library 21
17. WIKI Reference Library 22
18. Programs 22
19. Arcade 23
20. The Station MUD 24

PART IV. OFFICE 28
21. Office Fundamentals 28
22. Documents 28
23. Sheets 29
24. Presentations 29
25. Draw 30
26. Agenda 30
27. BASIC 30

PART V. PERSONALIZATION AND EVERYDAY USE 32
28. Appearance and Sound 32
29. Notifications 32
30. Installing the Station as an Application 33
31. Station Hours 33
32. Account Security 33

PART VI. STATION HOSTING AND ADMINISTRATION 35
33. Installation 35
34. Starting, Stopping, and the Station Manager 36
35. Network Access 37
36. Station Hours, Sleep, and Wake 37

37. SYSOP Control: Station Settings 38
38. Accounts, Permissions, and Moderation 39
39. Managing Modules and Content 40
40. Backup, Upgrade, and Maintenance 41

PART VII. DEVELOPER'S REFERENCE 42

41. Project Organization 42
42. Programs and Packages 42
43. Themes 43
44. MUD Worlds and Content 43
45. Testing, Building, and Further References 43

APPENDICES 45

Appendix A. Keyboard Reference 45
Appendix B. Quick Reference 46
Appendix C. Troubleshooting 46
Appendix D. Modules and Permissions 47
Appendix E. Glossary 48

PART I

Getting Started

1. Introducing StationNet

1.1 What a Station Is

A *station* is one running copy of StationNet together with everything it keeps: its name, its member accounts, its messages, and its files. The station runs on a single host computer, which may be a desktop computer, a laptop, a Raspberry Pi, or an Android phone. Other people reach the station over the local network with a web browser.

Every station has a name chosen by its operator and a *callsign*, a short code assigned when the station is first set up. Every member also receives a callsign. Callsigns appear beside usernames throughout the system, in the form `username - [CALLSIGN]`, so that members can be identified reliably.

One host computer may run several stations at once, each with its own name, network port, accounts, and content. The stations do not share information with one another.

1.2 Services and Modules

The facilities a station offers are called *services*. Each service is provided by a *module* that the SYSOP may turn on or off. Bulletin board, announcements, chat, mail, radio, files, the user directory, and programs are normally on. The imageboard, events, WIKI, videos, library, arcade, television, Office, and MUD are off until the SYSOP turns them on.

A service that is turned off does not appear in the navigation bar and cannot be opened. If this manual describes a service you cannot find, the station you are using does not offer it.

The SYSOP also supplies most of a station's content: the radio programming, the television channels, the books and videos, the games, and the MUD worlds. StationNet provides the means of presenting this material; it does not supply the material itself.

1.3 Classes of User

StationNet distinguishes three classes of user.

Guests are visitors who have not signed in. If the station permits guest access, a guest may read public services such as the bulletin board and announcements, but may not post, chat, or send mail.

Members are visitors who have signed in to an account. The status bar at the bottom of the screen shows a member as `MEMBER`.

SYSOPs are station operators. The person who sets up a station becomes its SYSOP and is shown as `SYSOP`. A SYSOP has two additional tabs, `SYSOP CONTROL` and `MODERATION`, and additional controls throughout the system. A SYSOP may also delegate particular duties to members, such as hosting radio broadcasts, creating events, moderating the imageboard, or building MUD worlds.

Parts I through V describe what a member can do. Operations reserved for SYSOPs or delegated members are identified as such.

1.4 What You Need

To use a station you need:

- A computer, tablet, or phone on the same network as the station.
- A current web browser, such as Chrome, Edge, Firefox, or Safari.
- The station's address, which the SYSOP will give you.
- An account on the station, unless you intend only to browse as a guest.

Some services require more. Live voice on the radio requires a microphone and, when you connect from a computer other than the host, a secure (`https://`) address. Installing the station as an application on a phone or desktop also requires a secure address. These requirements are described where they apply.

2. Connecting and Signing In

2.1 Opening the Station

Every station has an address of the form

```
http://<host-address>:<port>
```

For example, `http://192.168.1.20:8080`. The SYSOP will tell you the correct address. If you are sitting at the host computer itself, the address is usually `http://127.0.0.1:8080`.

Type the address into the address bar of your browser and press Enter. The station's welcome page appears, showing the station's name in large characters, its access policy, and, if the SYSOP has written one, a message about the station.

NOTE Bookmark the station's address, or install it as an application (see Chapter 30), so that you do not need to type it each time.

2.2 Registration Policies

A strip at the top of the sign-in page reports the station's registration policy:

PUBLIC ENROLLMENT OPEN	Anyone may create an account.
PRIVATE / INVITATION ONLY	A single-use invitation code from the SYSOP is required.
ENROLLMENT CLOSED	New accounts cannot be created. Ask the SYSOP for assistance.

2.3 Creating an Account

To create an account:

1. On the sign-in page, choose `NO ACCOUNT? CREATE ONE ->`, or on a private station `HAVE AN INVITATION? ENROLL ->`.
2. On a private station, enter the invitation code in the **Invitation code** field. If you followed an invitation link, the code is already filled in.
3. Enter a **Username** of 3 to 24 characters. Letters, numbers, the underscore (_), and the hyphen (-) are allowed.

4. Enter a **Password** of at least 10 characters, and enter it again in **Confirm password**.
5. Choose `CREATE ACCOUNT`.

The station creates the account, assigns you a random callsign, signs you in, and displays the confirmation `Account linked: <username> - [<callsign>]`. Usernames cannot be changed later, so choose carefully.

2.4 Signing In

1. Choose `LOGIN` or `> AUTHENTICATE`.
2. Enter your **Username** and **Password**.
3. If this is your own device, you may select **Remember me on this device**.
4. Choose `LOGIN`.

When you are signed in, the station displays `Link established. Welcome, <username>` and opens the STATION page.

With **Remember me on this device** selected, the station keeps you signed in on that browser across restarts of the browser and of the station until you disconnect. Without it, you are signed out when the browser is closed.

CAUTION	Do not select **Remember me on this device** on a computer that other people use.

2.5 If You Forget Your Password

StationNet does not use e-mail and cannot send you a reset link. Instead:

1. Attempt to sign in. After a failed attempt, a `FORGOT PASSWORD` button appears.
2. Make sure your username is entered, then choose `FORGOT PASSWORD`.
3. The station generates a temporary password and sends it privately to the station's SYSOPs. Contact a SYSOP, who will give it to you.
4. Sign in with the temporary password, then change it at once (Section 4.4).

2.6 Disconnecting

To sign out, open the `STATION` tab and choose `[ DISCONNECT ]` in the **CURRENT OPERATOR** panel. The station displays `Connection closed.` Disconnecting also ends any direct chats you have open and turns off your radio.

3. The Station Workspace

3.1 The Screen

Every station page has the same general arrangement.

The masthead. The station's name, drawn in large text characters, appears at the top. A short line of text, the *tagline*, may appear beneath it. The SYSOP sets both.

The station line. Beneath the masthead, a line shows `YOUR STATION`, the `SYSTEM DATE/TIME`, and several switches: `SFX`, which turns interface sounds on or off; `VFX`, which turns a screen scan-line effect on or off; and, where the station has a theme tune, `MUSIC`. These switches are described in Chapter 28.

If the station keeps regular hours, a second line shows them (Chapter 31).

The navigation bar. A row of tab names, separated by asterisks, leads to each service: `STATION`, `BULLETIN BOARD`, `ANNOUNCEMENTS`, `CHAT`, `MAIL`, `RADIO`, `FILES`, `USERS`, `PROGRAMS`, and others the station offers. The tab for the page you are on is highlighted. `CHAT` and `MAIL` show a count when something new is waiting.

The work area. The service you have chosen occupies the middle of the screen. Messages from the station appear above it, prefixed `SYS://`.

The radio player. When the radio is available, a small player is docked at the edge of every page. See Chapter 13.

The status bar. The bottom line shows the station callsign, your username and callsign, and your class (`MEMBER` or `SYSOP`). It also contains the link `GET YOUR OWN STATION`, which leads to a page of StationNet installation files if the SYSOP has provided them.

3.2 The STATION Page

The `STATION` tab opens the **Service Directory**. On the left is a numbered list of the station's services, each marked `* ONLINE`. Choose any entry to open that service.

On the right, the **CURRENT OPERATOR** panel shows your username and class and offers `[ PROFILE ]` and `[ DISCONNECT ]`. Beneath it, the **STATION ACTIVITY** panel summarizes recent events: the newest bulletin and reply, the latest chat message, the newest file and member, your unread mail, radio broadcasts now on the air, and current television listings. Each item is a link to the service concerned.

3.3 Conventions Common to All Services

StationNet services share a small number of conventions. Once learned, they apply everywhere.

Buttons are shown in capitals. A button ending in `+`, such as `NEW BULLETIN +`, creates something new.

Item menus. Items you are allowed to change, such as your own posts, carry a small downward triangle (▾). Choose it to reveal actions such as `DELETE`. StationNet asks you to confirm before anything is removed.

Reporting. Items written by other people carry a small ▾ actions menu. Open it and choose `REPORT`, then type a reason of 10 to 500 characters when asked to send that item to the SYSOPs for review. You may have one open report on any item and may file up to ten reports a day.

Times are displayed in your own device's time zone.

Keyboard navigation. When you are not typing in a field, pressing `/` moves the keyboard focus to the navigation bar (or, in the MUD, to the command line). Press Esc to release it. Individual services add their own keys; these are listed in Appendix A.

Guests. A guest who chooses a link that requires an account is sent to the sign-in page and, after signing in, is returned to the page requested.

4. Your Profile and Account

4.1 Viewing Your Profile

Your *profile* is the page other members see when they look you up. To open it, choose `[ PROFILE ]` on the STATION page, or choose your name in the user directory.

A profile shows whether the member is online, the time last active, the callsign, a short **Status** line, birthday and gender if supplied, the date the member joined, and an **About** section.

4.2 Editing Your Profile

1. Open your profile and choose `[ EDIT PROFILE ]`. (The `USERS` tab also has an `EDIT MY PROFILE` button.)
2. Complete any of the following fields. All are optional.
   - **Profile photo**: a PNG, JPEG, GIF, or WebP image. It is reduced to fit within 128 by 128 pixels and replaces any earlier photo.
   - **Status**: a line of up to 120 characters shown under your name.
   - **About me**: up to 2,000 characters.
   - **Birthday**: `MM/DD`, or `YYYY-MM-DD` to include the year.
   - **Gender**: up to 40 characters.
3. Choose `UPDATE MY PROFILE`.

4.3 Theme Preferences

If the SYSOP permits personal themes, the profile editing page also contains **Theme preferences**, where you may choose your own colours. See Section 28.1.

4.4 Changing Your Password

1. Open your profile and choose `[ EDIT PROFILE ]`.
2. Under **Change password**, enter your **Existing password**.
3. Enter a **New password** of at least 10 characters and confirm it.
4. Choose `CHANGE PASSWORD`.

Your StationNet password is separate from the Office password described in Chapter 21.

4.5 Deleting Your Account

The **Delete account** section at the foot of the profile editing page permanently removes your account and everything you have contributed: files, videos, images, Office documents, bulletins, feed posts and comments, chat messages, mail, events, radio uploads, saved program data, and MUD characters. Content in threads you started may also be removed.

To proceed, choose `DELETE ACCOUNT` and type your username. Choose `CONFIRM`, select the final `CONFIRM` box, and choose `DELETE`.

> **WARNING** Account deletion takes effect immediately and cannot be undone. The SYSOP cannot restore a deleted account.

The account that set up the station cannot be deleted in this way.

5. A First Session

This short session introduces the services most members use daily. It assumes you have an account and that the station offers the usual services.

1. **Sign in.** Open the station's address, choose `LOGIN`, and sign in as described in Section 2.4. The STATION page appears.
2. **Look around.** Read the **STATION ACTIVITY** panel to see what has happened recently. Choose `ANNOUNCEMENTS` to read any notices from the SYSOP.
3. **Complete your profile.** Choose `[ PROFILE ]`, then `[ EDIT PROFILE ]`. Write a short Status line, such as `New here - say hello`, and choose `UPDATE MY PROFILE`.
4. **Read the bulletin board.** Choose `BULLETIN BOARD`. Choose any subject to read it and its replies. To answer, type in **Post reply** and choose `SEND REPLY`.
5. **Say hello in chat.** Choose `CHAT`, type a greeting in the box at the bottom, and press Enter.
6. **Send a letter.** Choose `USERS`, find another member, and choose `MAIL`. Enter a subject and message and choose `DELIVER PRIVATE MAIL`.
7. **Turn on the radio.** In the radio player at the edge of the screen, choose `POWER: OFF` to switch it on, then choose `SCAN >` to find a station that is on the air.
8. **Disconnect.** When you have finished, open `STATION` and choose `[ DISCONNECT ]`.

PART II

Communication and Community

6. Users and Profiles

6.1 The User Directory

The USERS tab lists every registered member. Each entry shows the member's photo, ONLINE or OFFLINE, username, callsign, status line, and the time last active. SYSOPs are marked SYSOP.

To change the order, choose an option under **SORT BY**: **Most recently active**, **Newest registered**, **Oldest registered**, or **Alphabetical (A–Z)**. Then choose SORT. Long lists are divided into pages; use PREVIOUS, NEXT, or a page number.

Each entry offers PROFILE and MAIL. Members who are online also offer DIRECT CHAT.

6.2 Contacting Another Member

From another member's profile you may:

- Choose OPEN DIRECT CHAT, if the member is online (Section 7.2).
- Choose SEND PRIVATE MAIL (Chapter 8).
- Report the member's behaviour with REPORT USER in the profile's ▾ menu (Section 3.3).

6.3 User Feeds

Where the station offers user feeds, each profile has a **feed** of short posts beneath the profile details.

To post to your own feed, open your profile, type up to 200 characters in **Share a thought**, and choose POST TO YOUR FEED.

To react to a post, choose one of the five reaction buttons beneath it (✓, XD, <3, ;_;, >:c). You may hold one reaction per post; choosing another replaces it.

To comment, choose COMMENT beneath the post, type up to 500 characters, and choose COMMENT again. When you comment on another member's post, that member is notified by private mail. A profile shows the three newest comments on each post; choose VIEW ALL COMMENTS to see the rest.

You may delete your own posts and comments from their ▾ menus. The station limits the rate of posting: five posts in ten minutes, twenty comments an hour, and sixty reaction changes an hour.

7. Chat

7.1 Station Chat

Station chat is a single public conversation open to every member. Its messages are kept, so you can read what was said before you arrived.

1. Choose CHAT. The most recent messages are shown, and new ones appear automatically.
2. Type your message, up to 1,000 characters, in the box labelled **Transmit as**.

3. Press Enter, or choose `SEND`. Press Shift+Enter to begin a new line without sending.

To report a public message, use the report control shown on the message.

7.2 Direct Chat

Direct chat is a private, temporary conversation between two members who are both online.

To start a direct chat, choose `DIRECT CHAT` beside the member's name in the user directory, or `OPEN DIRECT CHAT` on their profile. Type and send messages as in station chat.

When someone writes to you, a count appears on the `CHAT` tab and an alert naming the sender appears near the top of every page. Choose the sender's name to open the conversation.

NOTE	Direct chat is not stored. The conversation is erased when either member disconnects, and it cannot be recovered later. Use private mail for anything you wish to keep.

If a SYSOP has muted your chat, the message `Your chat is muted` appears and messages cannot be sent until the restriction ends.

8. Private Mail

Private mail is a permanent, private message service between members of the station.

8.1 Reading Mail

Choose `MAIL`. The **Inbox** lists messages sent to you, with unread messages highlighted. A count on the `MAIL` tab shows how many messages are unread. Choose `SENT MAIL` to list messages you have sent.

Choose a message to open it. Messages between you and another member are shown together as one conversation, oldest first.

8.2 Sending Mail

1. Choose `COMPOSE +`. (Or choose `MAIL` beside a member's name.)
2. Select the **Recipient** from the list of registered members.
3. Enter a **Subject** of up to 120 characters.
4. Type the **Message**, up to 10,000 characters.
5. Choose `DELIVER PRIVATE MAIL`.

8.3 Replying and Deleting

To reply, open the conversation, type in the reply box at the bottom, and choose `SEND REPLY`. To remove a message from your own mailbox, choose `DELETE` beside it. Deleting a message does not remove it from the other member's mailbox.

The station also uses private mail for some notices: feed comments, SYSOP warnings, and password reset messages sent to SYSOPs.

9. Bulletin Board, Announcements, and Polls

9.1 The Bulletin Board

The bulletin board holds public discussions, called *bulletins*, each with its own replies. Choose `BULLETIN BOARD` to see the list. Each row gives the author, the subject with an excerpt, and the number of replies. A `POLL` tag marks a bulletin that includes a poll.

To read a bulletin, choose its subject.

To reply, type in **Post reply** at the foot of the bulletin and choose `SEND REPLY`.

To post a new bulletin:

1. Choose `NEW BULLETIN +`.
2. Enter a **Subject** (3 to 120 characters) and a **Message** (up to 5,000 characters).
3. Optionally, open `ADVANCED` to add a poll or set an automatic removal date (Sections 9.3 and 9.4).
4. Choose `POST`.

You may delete your own bulletins and replies from their ▾ menus. Deleting a bulletin also deletes all of its replies.

9.2 Announcements

The `ANNOUNCEMENTS` tab carries official notices from the SYSOP. Only SYSOPs may post announcements; everyone, including guests, may read them. An announcement may include a poll.

9.3 Polls

A poll is a question with up to ten responses, attached to a bulletin or announcement when it is first published.

To vote, select a response (or several, if the poll is marked `MULTIPLE VOTES ALLOWED`) and choose `VOTE`. Results are shown after you have voted. You may later choose `CHANGE VOTE` or `CLEAR VOTE`.

To add a poll when posting, open `ADVANCED`, select **Add a poll**, enter the **Question** and responses, and if desired select **Allow multiple votes**. The author of a poll may end voting with `CLOSE POLL`.

9.4 Automatic Removal

Under `ADVANCED`, the **AUTO-REMOVAL** fields let you choose a date and time at which a bulletin will be removed automatically. The time zone defaults to your device's own. Leave the date and time empty to keep the bulletin indefinitely. Bulletins with a removal date show `AUTO-REMOVES` and the time.

10. Imageboard

The imageboard is a single station-wide board of picture threads. It is available only to signed-in members.

10.1 Browsing

Choose `IMAGEBOARD` to see the **Thread catalog**. Each thread shows a preview image, subject, author, number of replies, and the time it was last *bumped* (that is, last replied to). Pinned threads appear first. Choose a thread's picture or subject to open it, and choose any picture in a thread to see it at full size.

10.2 Starting a Thread

1. Choose `NEW THREAD +`.
2. Enter a **Subject**. A **Message** is optional.
3. Choose an **Image**. PNG, JPEG, GIF, and WebP files of up to 10 MB are accepted. Animated images are shown by their first frame.
4. Choose `POST THREAD`.

10.3 Replying

Beneath the thread, type a **Message**, attach an **Image**, or both, and choose `POST REPLY`. A thread marked `LOCKED` accepts no replies.

10.4 Limits

The SYSOP sets the number of threads the board keeps. When a new thread would exceed the limit, the oldest threads are removed with their pictures. The SYSOP may also limit the number of pictures in each thread; after the limit is reached, text replies are still accepted.

SYSOPs, and members granted imageboard moderation, may pin, lock, and delete threads.

11. Events

The events calendar announces station activities and lets members say whether they will attend.

11.1 Viewing Events

Choose `EVENTS`. Upcoming events and events in progress are listed first, followed by past and cancelled events. Choose an event to see its description, start and end times in your own time zone, location, link, a countdown, and the members who are going or interested.

11.2 Responding

On the event page, choose `GOING` or `INTERESTED`. The chosen button is marked with a tick. To change your mind, choose the other button, or `WITHDRAW`.

11.3 Creating an Event

The SYSOP decides who may create events: SYSOPs only, members of selected groups, or all members. If you are permitted, `CREATE EVENT +` appears on the EVENTS page.

1. Choose `CREATE EVENT +`.
2. Enter a **Name** and **Description**.
3. Enter the **Start date** and **Start time**, and optionally the end date and time.
4. Check the **Timezone**. It defaults to your device's time zone and is given in the standard form `Region/City`, for example `America/Edmonton`.

5. Optionally enter a **Location** and a **URL**.
6. Choose `CREATE EVENT`.

The organizer may later choose `EDIT` or `CANCEL EVENT`. A cancelled event remains listed so that members know it will not take place.

12. File Archive

The file archive is the station's shared collection of downloadable files. Choose `FILES` to open it.

12.1 Finding and Downloading Files

The archive is divided into five categories: **General**, **Documents**, **Images**, **Software**, and **Other**. Choose `ALL` or a category name to filter the list. Each entry shows the category, size, name, description, uploader, and date. Picture files show a small preview.

Choose `DOWNLOAD` to save a file to your device.

12.2 Uploading Files

1. Choose `UPLOAD FILE +`.
2. Choose the **File**, select a **Category**, and add a **Description** of up to 500 characters.
3. Choose `UPLOAD TO ARCHIVE`.

The heading of the archive shows how much of the station's total storage is in use, and the upload page shows the largest file accepted. The SYSOP may limit uploading to SYSOPs or turn it off entirely, in which case `UPLOAD FILE +` does not appear.

You may delete files you uploaded from their ▾ menus.

CAUTION	Files in the archive are visible to every member. Do not upload private documents or material you do not have the right to share.

PART III

Information and Entertainment

13. Radio

The StationNet radio is a shared receiver and transmitter. Members tune to frequencies, measured in megahertz (MHz), to hear broadcasts and to talk with one another by voice, Morse code, or still pictures. The radio is an internal facility of the station; it does not use real radio equipment.

13.1 The Radio Player and the Radio Console

The radio can be operated in two places.

The **radio player** is docked on every page except the radio console. It shows the power switch, the frequency, and what is playing. Choose `R` to expand it and reveal tuning, scanning, volume, `MUTE`, and `FX` controls, and `OPEN RADIO`. Because the player is present on every page, you may listen while you use other services.

The **radio console**, opened with the `RADIO` tab or `OPEN RADIO`, offers the complete set of controls: a frequency display, a band selector, a live signal monitor, the **Station broadcast directory**, and controls for transmitting.

13.2 Listening

1. Choose `POWER: OFF`. The switch changes to `POWER: ON`.
2. Tune a frequency in any of these ways:
   - Type a frequency, such as `7.25`, in the **Frequency MHz** field and choose `TUNE` (or press Enter).
   - Choose `TUNE - 0.50` or `TUNE + 0.50` to step down or up.
   - Choose a **Band** to move to it.
   - Choose `< SCAN` or `SCAN >` to search for the next active frequency.
   - Choose an entry in the **Station broadcast directory**.
3. Adjust **Volume**, or choose `MUTE`.

The display shows the name and message of the broadcast, the current title and artist, and the band. `FX` adds an analogue receiver sound to what you hear; choose it again to hear the plain signal.

NOTE	Browsers will not play sound until you have clicked or tapped on the page. If the radio is on but silent, choose any control on the page.

13.3 Bands

The frequency range is divided into *bands*, each set aside for a particular purpose and kind of transmission. A new station has the following bands; the SYSOP may change them.

Band	Range (MHz)	Use
BROADCASTS	7.00–13.99	Hosted programmes

Band	Range (MHz)	Use
MORSE / CW	14.00–27.99	Morse code
DATA TRANSMISSIONS	28.00–144.99	Still pictures (SSTV)
GENERAL COMMUNICATION	145.00–432.99	Live voice
EXPERIMENTAL / UNRESTRICTED	433.00–999.99	Morse and pictures

Whether you may transmit on a band depends on the station's radio permissions: `FULL` allows receiving and transmitting, `LISTEN` allows receiving only, and `DENIED` allows neither. Only one person may transmit on a frequency at a time.

13.4 Transmitting

When you tune to a band on which you may transmit, the console shows the controls for that band. If the band supports more than one kind of transmission, choose one under **Active mode**.

Live voice. Hold `HOLD TO TALK [V]`, or hold the V key while you are not typing in a field, and speak. Release to stop transmitting and clear the frequency for others. The browser will ask permission to use the microphone the first time. Voice travels directly between browsers and is never recorded.

NOTE Browsers permit microphone use only on secure connections. Live voice works at the host computer itself, and from other computers only when the station is reached through an `https://` address. The console reports this condition if it applies.

Morse code (CW). Choose `BEGIN CW TRANSMISSION`, then hold `HOLD KEY [SPACE]` (or the Space bar) for each dot and dash. **Tone Hz** sets the pitch and **Speed WPM** the speed in words per minute. Select **CW decoder** to have received Morse shown as text. A **MORSE REFERENCE** table is provided.

Pictures (SSTV). Choose a **Local image**, check the preview, and choose `TRANSMIT PICTURE`. Pictures are sent at a small size (80 by 60 pixels) line by line. A picture you receive builds up in the **RX PICTURE** box; choose `SAVE RECEIVED` to keep it.

13.5 Hosted Broadcasts

Members whom the SYSOP has authorized to host broadcasts see a `MANAGE BROADCASTS` button and a **Broadcast console** on the radio console. A host uploads audio files (MP3, OGG, WAV, or MIDI), arranges them in playlists or in *programmes* that mix music with announcements, Morse, pictures, and live microphone breaks, and starts the broadcast on an assigned frequency.

During a broadcast the host may choose `TAKE CONTROL` to act as disc jockey, then `PAUSE`, `SKIP`, `HOLD TO TALK`, or `PLAY NOW` for prepared content. Listeners who tune in midway join at the current point of the programme.

14. Television

The `TV` tab opens the station's television set. Television is *linear*: each channel follows a schedule set by the SYSOP, and when you tune in you join whatever programme is on the air at that moment, as with broadcast television.

14.1 Watching

1. Choose `TV`.
2. Choose `POWER` on the screen (or the ⏻ key on the receiver panel).
3. The set tunes to the station's default channel, or to the channel you last watched.

The receiver panel beside the screen provides channel up and down (`CH ▲ ▼`), volume (`VOL − +`), `MUTE`, `GUIDE`, `INFO`, fullscreen, and a numeric keypad for entering a channel number directly.

`INFO` displays the channel, the current programme with its progress, a description, and the next programme. A channel with nothing scheduled shows an `OFF AIR` test card with the time the channel resumes.

14.2 The Program Guide

Choose `GUIDE` (or press G) to display the program guide, a grid of all channels against time. Use the arrow keys to move through it and choose a programme, or press Enter, to read about it or tune to it. The guide's own buttons are `Page Up`, `Page Down`, `+24 Hours`, and `−24 Hours`. Times are shown in your local time.

14.3 Keyboard Controls

Key	Action
0–9	Enter a channel number
↑ ↓, Page Up, Page Down	Change channel
[and]	Volume down and up
M	Mute
G	Guide
I	Information
F	Fullscreen
Esc	Close the guide or information

15. Videos

The `VIDEOS` tab contains video recordings that can be watched at any time.

To find a video, type part of its title in **Search videos by title** and choose `SEARCH`, or browse the pages of the catalogue. Choose a video to open it. Use the player's own controls to play, pause, seek, change volume, and view fullscreen.

The SYSOP runs the video service in one of two modes. In **Community** mode, members may upload videos and comment on them. In **Curated** mode, only the station publishes videos, authors are not shown, and comments are not offered.

To upload a video (Community mode):

1. Choose `UPLOAD VIDEO +`.
2. Enter a **Title** and, optionally, a **Description**.

3. Choose a video file in MP4, WebM, or Ogg format, within the maximum size shown on the upload form.
4. Choose `UPLOAD VIDEO`.

The SYSOP sets the maximum size of each video separately from each member's total storage quota. Member uploads must fit both limits; station uploads use only the file size limit. The default file limit is 120 MB. Sizes use 1 MB = 1,048,576 bytes. Interrupted uploads must be restarted. **To comment**, type in **Add a comment** beneath the video and choose `POST COMMENT`.

16. Library

The `LIBRARY` tab holds books in EPUB format and documents in PDF format.

16.1 Finding a Book

Use the controls at the top of the catalogue: **Search titles, authors, and descriptions**, a **Category**, a **Format** (EPUB or PDF), and **Sort by**. Choose `APPLY`; `CLEAR` restores the full catalogue. Choose a cover to see the book's details.

16.2 Reading

On the details page, choose `READ BOOK` (EPUB) or `READ PDF`. If you have read the book before, the button reads `RESUME READING` and returns you to where you stopped. When the SYSOP permits it, a `DOWNLOAD` button saves a copy of the file.

EPUB books are displayed one chapter at a time in the station's colours. Select a **Chapter** from the list, or choose `PREVIOUS CHAPTER` and `NEXT CHAPTER`. Choose `A−` and `A+` to change the size of the text.

PDF documents keep their original appearance. Choose ◀ and ▶ (or press the Left and Right arrow keys) to turn pages, or type a page number. Choose − and + to zoom, or `FIT WIDTH`.

16.3 Bookmarks and Reading Position

When you are signed in, your reading position is saved automatically and follows you to any device you use on the same station.

To set a bookmark, choose `BOOKMARK` while reading. **To return to a bookmark**, open **Bookmarks** above the text and choose the bookmark. **To remove one**, choose × beside it. Guests may read, but positions and bookmarks are not saved for them.

16.4 Contributing Books

In Community mode, members may choose `UPLOAD EPUB / PDF +` to add a book. Title, author, and description are read from the file where available, and may be corrected afterwards with `EDIT DETAILS`. Uploads count against a quota set by the SYSOP.

17. WIKI Reference Library

The WIKI tab opens the station's **REFERENCE LIBRARY**: offline copies of encyclopedias, dictionaries, manuals, and other reference works that the SYSOP has installed. Each work is called a *collection*. Because the collections are stored on the station itself, they are available without an Internet connection.

17.1 Browsing and Searching

The opening page lists the available collections. Choose OPEN COLLECTION → to go to a collection's main page, and follow links within its articles as you would on any web site.

To search, select a collection in the search bar, type words in **Find articles and pages**, and choose SEARCH. Choose a result to open it.

While reading, use ← BACK and FORWARD → to retrace your steps, the **COLLECTION** list to switch collections, and ALL COLLECTIONS to return to the list.

If the page reports LIBRARY UNAVAILABLE, no collection is installed or the reference reader could not start. Inform the SYSOP.

18. Programs

The PROGRAMS tab opens the **Program Manager**, a directory of software that runs within StationNet: games, utilities, creative tools, and productivity programs.

18.1 Finding and Launching a Program

1. Choose PROGRAMS.
2. Narrow the list by typing in **FIND PROGRAM**, or choose a category: GAMES, UTILITIES, CREATIVE, EDUCATIONAL, PRODUCTIVITY, OTHER, and, when Office is enabled, OFFICE.
3. Choose a program to see its details: version, author, the devices it supports, and the input it requires.
4. Choose LAUNCH PROGRAM >.

The list may also be operated from the keyboard: press ↑ and ↓ to select a program and Enter to open it.

If your device lacks something the program needs, such as a keyboard, the program pauses with an explanation. You may choose **Launch anyway** to try it regardless.

18.2 Pinned and Recent Programs

Choose PIN TO FAVORITES on a program's details to add it to the **PINNED** list at the left of the Program Manager. Programs you have launched appear under **RECENTLY USED**. Choose UNPIN to remove a pinned program.

18.3 Running Programs

Programs open within the StationNet page. Some fill the whole page in *immersive* presentation; choose **Exit immersive**, or press Esc, to return. Many programs can also be shown fullscreen.

Programs save their own information, such as high scores and documents, to your account on the station. Some keep shared station-wide records, such as a community score table.

18.4 Programs Supplied with StationNet

StationNet is supplied with a set of built-in programs. A station may disable any of them or add others.

Program	Category	Description
Calculator	Utilities	Standard and scientific calculator with memory
Clock	Utilities	World clocks, alarms, timers, and stopwatch
Sleep Calculator	Utilities	Bedtimes and waking times by sleep cycle
Focus Console	Productivity	Work timer with session log and statistics
Personal Calendar	Productivity	Private month calendar and daily planner
ASCII/ANSI Art Studio	Creative	Coloured character art
MIDI Composer	Creative	Multi-track music composition and MIDI export
Breakout	Games	Brick-breaking game
Snake	Games	Classic, Sprint, and Maze Snake
Number Guess	Games	Number-guessing game
Light Cycles	Games	Light cycle arena game
Cybercrawler	Games	Turn-based grid exploration
Echo Choir: Remix	Games	Musical puzzles
Oddling Garden	Games	Virtual pets
Void Trader	Games	Space trading simulation
Wirestorm Rally	Games	Off-road racing

19. Arcade

The `ARCADE` tab offers games for classic home consoles and arcade machines, played in the browser through emulation. The SYSOP or the members supply the game files; StationNet supplies the emulators.

19.1 Finding a Game

Use **Search games**, **Sort by**, and the **Platforms** and **Genres** menus, then choose `APPLY FILTERS`. Choose a game's cover to open its page, which lists its platform, genre, year, developer, and other details.

Supported platforms include the Nintendo Entertainment System, Super Nintendo, Game Boy, Game Boy Color, Game Boy Advance, Nintendo 64, Nintendo DS, Sega Genesis/Mega Drive, Master System, Game Gear, Atari 2600, 7800, and Lynx, TurboGrafx-16, ColecoVision, MAME arcade sets, and Sony PSP.

19.2 Playing

Choose ▶ `PLAY <title>`. The emulator loads and starts the game. The emulator's own menu bar, at the foot of the game area, provides controls, volume, fullscreen, and the emulator's settings, including the keyboard and gamepad assignments.

NOTE The station does not keep emulator save states for you. If you use the emulator's save-state feature, keep any file it gives you.

Sony PSP games require a secure (`https://`) connection or play at the host computer, and a browser with WebGL 2 support. Very large games may exceed the memory available to a browser on small devices.

19.3 Uploading Games

When the arcade is in Community mode, members may choose `UPLOAD GAME +`, select the platform, and supply the game file, cover picture, and details. Uploads count against a quota.

CAUTION	Upload only game files you are entitled to share.

20. The Station MUD

The MUD is a text-based, multiplayer role-playing game. You explore a world described in words, meet other players and the game's characters (called NPCs), take on quests, gather possessions, and fight opponents. A station may run several worlds, and each world is a separate game.

20.1 Choosing a World and Creating a Character

Choose `MUD`. The **Choose a world** page lists the worlds the station offers. Each world requires its own character; your progress, skills, possessions, and coins remain in the world where they were earned.

1. Choose `CREATE CHARACTER` beside a world.
2. Work through the steps, choosing `NEXT` and `BACK` as required:
    1. **Identity**: a character name of 2 to 32 letters. Spaces, apostrophes, periods, and hyphens are permitted. Other players see this name, which is separate from your StationNet username.
    2. **Race**: each race lists its traits and attribute adjustments.
    3. **Archetype**: your character's calling, with its own bonuses.
    4. **Attributes**: distribute the points shown among Strength, Agility, Intellect, and Vitality.
    5. **Skills**: select the number of starting skills shown.
    6. **Faction**: offered only in worlds that have factions.
    7. **Review**.
3. Choose `CREATE CHARACTER`.

Your choices cannot be changed later except by deleting the character with `DELETE CHARACTER` and beginning again. To return to a world, choose `CONTINUE`. To switch worlds, choose `CHANGE WORLD`.

20.2 The MUD Screen

The **GAME** panel carries a running account of what happens around you. At its top are the name of your current location and three buttons: `ROOM` shows the full description of the location with its exits, items, NPCs, enemies, and other players; `LATEST` returns to the newest activity; and `CLEAR` empties the account.

Below it is the **command line**. At the side, panels show your **Character** (level, health, mana, experience, attributes), **Inventory**, **Quests**, and, in some worlds, faction **Standing**. Choose a panel's title to open or close it.

20.3 Entering Commands

Type in the command line and press Enter or choose `SEND`.

- Text without a leading slash is speech, heard by everyone in the same location.
- Text beginning with `/` is a command, such as `/look`.
- While you type a command, suggestions appear; press Tab to accept one.
- Press ↑ and ↓ to recall earlier commands.
- Enter `/help` for a list of commands, or `/help <command>` for details of one.

Where a name contains spaces, enclose it in quotation marks: `/get "Red Apple" from "Oak Chest"`.

20.4 Moving About

Each location, or *room*, has exits leading to other rooms. `/exits` lists them.

```
/go north        Leave by the north exit
/north  or  /n   The same, in short form
/look            Describe the room again
/map             Show a map of the surrounding area
```

The directions `north`, `south`, `east`, `west`, `northeast`, `northwest`, `southeast`, `southwest`, `up`, and `down` may each be used as a command or abbreviated (`/n`, `/ne`, `/u`, and so on). Some exits have names instead of directions; use `/go <exit name>`.

In open country, `/travel <direction>` walks steadily in one direction, one room at a time, until something of interest occurs; `/stop` halts the journey. `/directions <place>` gives the way to a place you know, and `/destinations` lists the places you know.

20.5 Talking to Players and Characters

Command	Purpose
`/say <text>`	Speak to the room (plain text does the same)
`/tell "<name>" <text>`	Speak privately to one player
`/reply <text>`	Answer the last private message
`/emote <action>`	Describe an action
`/who`	List players in the world
`/talk <npc>`	Begin a conversation with an NPC
`/respond <number>`	Choose a reply in a conversation
`/block <name>`	Ignore a player

20.6 Possessions

`/inventory` (or `/i`) lists what you carry and wear. `/get` and `/drop` pick up and put down items; `/open`, `/close`, and `/put` work with containers; `/equip` and `/unequip` wear or remove gear; `/use` uses or reads an item. Where a merchant is present, `/shop` shows what can be bought and sold.

20.7 Quests

NPCs may offer quests. `/quests` lists your active quests and their objectives, which are also shown in the **Quests** panel. Many worlds use `/quest available`, `/quest accept`, and `/quest turnin` to find, accept, and complete quests.

20.8 Combat and Recovery

Some NPCs are hostile. `/attack <enemy>` attacks; `/defend` reduces the damage you take and protects your companions; `/flee` attempts escape; `/cast <spell>` uses a known spell (see `/spells`); `/combat` reports the state of the fight. Outside combat, `/rest` restores part of your health and mana.

If your character is defeated, it recovers at the starting location with half its health. In ordinary worlds, part of the experience gained toward the next level is lost, but levels are never lost.

As your character gains experience it rises in level and earns *attribute points*, spent with `/train`, and *skill points*, used with `/learn` and `/practice`. `/character` shows your full statistics and `/skills` your skills.

20.9 Dungeons and Expeditions

Some worlds contain dungeons that are generated afresh for each visit. Where the station offers the procedural dungeon world, `/expedition` starts a random expedition and `/daily` enters the station-wide Dungeon of the Day. Other players may join an expedition with `/join <code>`. `/run` shows its progress, `/complete` finishes it from the exit, and `/abandon` leaves at once.

NOTE The commands available depend on the world. Many worlds add banking, housing, professions, vehicles, and other systems. `/help` always lists the commands that apply where you are.

20.10 Building Worlds with World Forge

Builders and the SYSOP create and edit worlds with **World Forge**, linked as `WORLD FORGE` in the navigation bar and from each world's MUD page. The first screen lists the station's installed worlds and any new worlds not yet installed. `Create a world` starts a world with one arrival room; `Import world ZIP` opens a world package as a new draft.

Drafts. Opening a world edits its *draft*, a copy kept on the station and shared by every builder. Each change is saved to the draft at once; `UNDO` and `REDO` step through recent changes, and `CHECKPOINT` (Ctrl+S) records a backup that **Backups & recovery** can restore. Players see nothing until the draft is published. Builders editing the same draft are named in the header.

Tabs.

- **World map.** The *Grid* view places rooms on numbered squares, draws passages, and paints wilderness terrain with the brush, rectangle, and ellipse tools. Layers hold upper and lower floors, joined by `up` and `down` passages. The *Connections* view diagrams rooms that have no grid position.
- **Content library.** Every kind of content: items, characters, dialogue, quests, factions, shops, professions, housing, vehicles, transit, triggers, dungeon templates, terrain presets, races, classes,

skills, abilities, and effects. Selecting a record opens its guided editor; the *JSON* tab edits it directly. *Used by* lists what refers to it, and records still in use cannot be deleted.

- **Review.** Checks the draft for errors before it is published.
- **Handbook.** The creator handbook: every field, with examples.
- **Live world.** Acts on the running world: who is where, property holdings, vehicles, script state, and builder access. These actions take effect at once and do not change the draft.

Publishing. `PUBLISH` checks the draft and shows what will change. A draft that only changes text and numbers is applied in place, and players notice nothing but the new text. Adding, removing, or reconnecting content upgrades the world: characters keep their levels, quests, reputation, and possessions; characters standing in a removed room move to the starting room; items lying in rooms return to their authored places; and dungeon runs in progress end. A new world is *installed* hidden; the SYSOP enables it under **MUD service → Worlds**. If a world is upgraded from Settings while a draft is open, the draft is marked out of date and may be rebased from **Backups & recovery**.

NOTE The home world may be edited and published like any other. Take a station backup (Section 40.1) before its first full publish, which converts the home world to the package format.

PART IV

Office

21. Office Fundamentals

StationNet Office is a set of six private productivity programs: **Documents**, **Sheets**, **Presentations**, **Draw**, **Agenda**, and **BASIC**. Open them with the `OFFICE` tab, or from the `OFFICE` category in the Program Manager. Office is available only to signed-in members.

21.1 The Office Password

Office work is *encrypted* in your browser before it is sent to the station. Neither the SYSOP nor anyone else can read it. The encryption is controlled by a single **OFFICE Password**, which is separate from your StationNet password.

The first time you open any Office program, you are asked to choose an OFFICE Password of at least 12 characters and confirm it.

Thereafter, each program opens with an **Unlock** panel:

1. Enter your **OFFICE Password**.
2. Optionally select **Remember on this browser**, so that the programs stay unlocked on this browser until you choose `LOCK` or disconnect.
3. Choose the unlock button.

Unlocking one Office program unlocks all of them in the same browser tab. Choose `LOCK` in any program when you leave your desk.

> **WARNING** The OFFICE Password is not stored anywhere and cannot be reset by the SYSOP. If you forget it, everything you have saved in Office becomes permanently unreadable.

21.2 Operations Common to All Office Programs

Each program keeps a private list of your files, and each provides `NEW`, `SAVE`, `DELETE`, `UNDO`, and `REDO` where they apply.

Documents, Sheets, Agenda, and BASIC save your work automatically shortly after you stop typing, and report the save status on screen. In Presentations and Draw, choose `SAVE` to save. If the same file is open in two browser tabs, StationNet prevents the older tab from overwriting newer work.

Import and **export** take place entirely in your browser. Importing creates a new Office file from a file on your device; exporting saves a copy of an Office file to your device in a common format. Exported copies are not encrypted.

22. Documents

Documents is a word processor.

Writing and formatting. Select a paragraph style (Normal, Title, Subtitle, Heading 1 to 3, Quote, or Code) and use the toolbar for bold, italic, alignment (`LEFT`, `CENTER`, `RIGHT`, `JUSTIFY`), bulleted and numbered lists, indentation, links, horizontal rules, images, and tables (`TABLE`, `+ ROW`, `+ COL`). `FIND` opens find and replace.

Importing. Choose `IMPORT FILE` and select a Word (`.docx`), OpenDocument (`.odt`), text, Markdown, HTML, or RTF file.

Exporting. Choose `EXPORT`, then `WORD .DOCX`, `ODT`, `PDF`, `HTML`, `MARKDOWN`, or `PLAIN TEXT`.

Printing. Choose `PRINT`. `PRINT THEME` previews the page as it will print, in black on white.

NOTE Word and OpenDocument files keep their general structure, but complex page layouts, tracked changes, comments, and unusual fonts may be lost on import or export. Markdown and plain text lose page styling; plain text also loses pictures.

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23. Sheets

Sheets is a spreadsheet program. Each workbook may contain several sheets.

Entering data. Select a cell and type a value or a formula. Formulas begin with =, for example `=SUM(B2:B10)`. Set number formats (`GENERAL`, `NUMBER`, `CURRENCY`, `PERCENT`, `DATE`) and alignment from the toolbar.

Arranging data. Use `+ ROW`, `− ROW`, `+ COL`, and `− COL` to insert and remove rows and columns; `FILL →` and `FILL ↓` to copy across or down; `SORT ↑` and `SORT ↓` to sort; `FILTER` to show only matching rows; and `FREEZE` to keep headings in view. `RENAME` and `DELETE SHEET` act on the current sheet.

Charts. Select a range and choose `BAR CHART`, `LINE CHART`, or `PIE CHART`.

Importing and exporting. `IMPORT FILE` accepts Excel (`.xlsx`, `.xls`), OpenDocument (`.ods`), CSV, and TSV files. For text files you may choose the character encoding and separator. `EXPORT` offers `EXCEL .XLSX`, `ODS`, `CSV`, `TSV`, and `PDF / PRINT`.

24. Presentations

Presentations prepares and shows slide presentations.

Building slides. Choose a layout (`BLANK`, `TITLE`, `TITLE + BODY`, `TWO COLUMNS`) and a theme (`STATION`, `LIGHT`, `DARK`, `WARM`). Add objects with `+ TEXT`, `+ IMAGE`, `+ RECT`, `+ ELLIPSE`, `+ LINE`, and `+ TABLE`. Selected objects may be moved, resized, aligned, sent to the `FRONT` or `BACK`, duplicated (`DUP`), or deleted. Each slide may have a background picture and a transition (`NONE` or `FADE`).

Presenting. Choose `PRESENT`. Advance with the Right arrow, Page Down, or the Space bar; go back with the Left arrow; press Esc or choose `EXIT` to finish.

Importing and exporting. `IMPORT .PPTX` opens a PowerPoint file. `EXPORT .PPTX`, `PDF`, and `SLIDE .PNG` save the presentation or the current slide.

Remember to choose `SAVE`; Presentations does not save automatically.

25. Draw

Draw produces diagrams, flowcharts, posters, and simple page layouts.

Select a tool: SELECT, PEN (freehand), LINE, ARROW, CONNECTOR, RECT, ELLIPSE, TRIANGLE, DIAMOND, TEXT, or IMAGE. Drag on the page to draw. Connectors attach to shapes and follow them when the shapes are moved.

With objects selected, GROUP and UNGROUP combine and separate them, FRONT, FORWARD, BACKWARD, and BACK change their stacking, and DUPLICATE and DELETE copy and remove them. The zoom list sets the view from 50% to 200%, or FIT.

To export, choose PNG for a picture or SVG for a scalable drawing. Choose SAVE to keep your work in Office.

26. Agenda

Agenda combines a calendar, a task list, and an address book.

Calendar. Choose MONTH, WEEK, or DAY, and use ‹, ›, and TODAY to move through time. Choose + EVENT, enter the details, and choose SAVE EVENT. Events may repeat.

Tasks. Choose TASKS, then + TASK. Tasks may have subtasks (+ SUBTASK). Choose SAVE TASK.

Contacts. Choose CONTACTS, then + CONTACT, and SAVE CONTACT.

Reminders. Choose ENABLE REMINDERS and allow notifications when your browser asks. Reminders are given only while Agenda is open and unlocked in a browser tab.

27. BASIC

BASIC is a programming environment for the BASIC language, compatible with programs written for QBasic and QB64. Programs can draw graphics, play sound, and respond to the keyboard, mouse, and touch screen.

27.1 Writing and Running a Program

1. Choose NEW.
2. Type the program in the CODE view. For example:

```
CLS
INPUT "What is your name"; N$
PRINT "Hello, "; N$
```

3. Choose RUN ▶. The program runs in the SCREEN view.
4. Choose STOP ■ to end a program that is still running.

Error messages appear in the console beneath the screen. To study working programs, select one from **Open an example...**.

Each run takes place in a closed area of the browser that cannot reach the station, your Office files, or the network.

27.2 Importing and Exporting

`IMPORT .BAS` opens a BASIC source file from your device. `EXPORT…` offers two forms:

- **BASIC file (.bas)**: the program text.
- **StationNet Package (.zip)**: an installable program. Fill in the name, description, category, and device requirements. Give the file to the SYSOP, who can install it in the Program Manager for all members.

PART V

Personalization and Everyday Use

28. Appearance and Sound

28.1 Themes

StationNet's appearance is governed by two colour schemes: the **site-wide theme**, which colours pages, panels, and controls, and the **masthead theme**, which colours the station name at the top of the screen. The SYSOP chooses the station's defaults. Supplied themes include Default/Cyan, Amber, Terminal Green, Bright Green, Magenta, Electric Blue, Crimson, Violet, Monochrome, and Vintage Terminal. Themes added by the SYSOP are marked `[CUSTOM]`.

To choose your own:

1. Open your profile and choose `[ EDIT PROFILE ]`.
2. Under **Theme preferences**, choose a **Site-wide theme** and a **Masthead theme**. The preview panels show the effect. Choose **Station default** to follow the station.
3. Choose `SAVE THEME PREFERENCES`.

If **Theme preferences** does not appear, the SYSOP has required all members to use the station's themes.

28.2 Interface Switches

The switches on the station line apply to the browser you are using and are remembered by it.

SFX	Interface sound effects, such as clicks and the alerts for new mail and direct chat.
VFX	A scan-line effect resembling a cathode-ray screen.
MUSIC	The station theme tune, if the SYSOP has provided one. Your choice is saved with your account.

Where supported, a fullscreen switch is also shown.

28.3 Accessibility

All pages can be operated from the keyboard. A **Skip to content** link at the top of each page moves directly to the work area. Moving tagline effects are stopped when your device is set to reduce motion. Text size follows your browser's zoom setting (Ctrl and + or −).

29. Notifications

StationNet notifies you within the station's pages:

- A count on the `MAIL` tab shows unread mail, with a sound if `SFX` is on.
- A count on the `CHAT` tab and an alert at the top of the page announce a direct chat message. The alert clears when you open the conversation.

- The **STATION ACTIVITY** panel summarizes new bulletins, files, members, broadcasts, and television.
- Agenda reminders use your browser's notifications (Chapter 26).

Notifications appear only while a station page is open in your browser.

30. Installing the Station as an Application

A station can be installed on a phone, tablet, or computer so that it opens in its own window with its own icon, like any other application.

1. Open the station in a browser that supports installation, such as Chrome or Edge, or Safari on an iPhone or iPad.
2. Choose the browser's **Install** or **Add to Home Screen** command.

The installed application uses the station's name and, if the SYSOP has chosen, its icon. A connection to the station is still needed; when the station cannot be reached, the application shows **Station unavailable**.

NOTE Browsers offer installation only for secure (`https://`) addresses and for the host computer itself. A station reached through a plain `http://` network address can be used in the browser but cannot be installed.

On a phone, all services can be used in the browser. Some programs and games show on-screen touch controls.

31. Station Hours

Some stations are open only at certain times. When a station keeps hours, a line beneath the station line shows them, for example `STATION HOURS: 8:00AM - 9:00PM`. Other messages that may appear there are `HOURS TEMPORARILY EXTENDED` and `OPEN EARLY`.

When the station closes, it stops, and pages no longer respond until it opens again. Save your work and finish conversations before closing time.

32. Account Security

Observe the following practices:

- Choose a password of at least ten characters that you do not use elsewhere.
- Use **Remember me on this device** and Office's **Remember on this browser** only on your own devices.
- On a shared computer, choose `LOCK` in Office and `[ DISCONNECT ]` when you finish.
- If you suspect someone else has used your account, change your password at once (Section 4.4) and inform the SYSOP.

SYSOPs can read the station's public services and administer accounts. They cannot read your Office files. Private mail and chat are stored on the station computer and should not be regarded as secret from the person who operates it.

PART VI

Station Hosting and Administration

This part is for the SYSOP. It describes how to install StationNet, start and operate stations, and administer accounts and content.

33. Installation

33.1 The Software, the Station, and Its Data

Three things should be kept distinct:

- **StationNet** is the software: the program and its supplied files. It never writes to its own folder.
- A **station** is one configured, running service with its own name and port.
- A station's **data** is kept in its own folder under `stations/`. Every account, message, upload, installed program, theme, world, and reference collection belongs to the station folder.

A typical installation is arranged as follows:

```
StationNet/
  run.py, stationnet/, .venv/   the software (packaged: core/)
  stations/
    default/                    one folder per station
      station.json              name, port, network access, hours
      station.db                the station database
      uploads/, themes/, ...    files, programs, worlds, WIKI data
      run/                      status and log while running
```

33.2 Requirements

The host computer may run Windows, Linux, macOS, or Raspberry Pi OS. Source installations require Python 3.11 or newer. Packaged Windows builds include everything they need. Packaged Linux and Raspberry Pi builds require Python 3.11 with venv. The Station Manager requires Tkinter (`python3-tk` on Debian and Raspberry Pi OS). StationNet uses no separate database server.

33.3 Installing a Packaged Release

Windows. Unzip the package and double-click `install.bat`. The software is installed to `%LOCALAPPDATA%\StationNet`. Start a station with `start.bat` in that folder, or manage several with `manager.bat`. To run without installing, run `start.bat` from the unzipped folder.

Linux and Raspberry Pi. Unzip the package for your platform and run:

```
sh install.sh
~/.local/share/stationnet/start.sh
```

The Station Manager is `~/.local/share/stationnet/manager.sh`. Packaged releases install without network access.

Android. The StationNet Android application hosts a station on a 64-bit ARM phone running Android 8 or later. Tap **Start station**, complete the first-run setup in the phone's browser, then enable the phone's hotspot so that nearby devices can connect to the addresses shown. The Android application

does not yet include the WIKI reader.

33.4 Installing from Source

Windows: double-click `install.bat`, then start with `run.bat` (one station) or `manager.bat` (several stations).

Linux and macOS:

```
chmod +x install.sh run.sh manager.sh uninstall.sh
./install.sh
./run.sh
```

Source installation downloads Python packages and the WIKI reader.

33.5 First-Run Setup

When a new station starts for the first time, open `http://127.0.0.1:8080` on the host computer. The **Initialize local station** page asks for:

- The **Station name** (2 to 80 characters) and **Station description**.
- The **Administrator username** and **Password** of the first SYSOP.
- The **Registration policy**: Open, Private, or Disabled.

Choose `INITIALIZE STATION`. The station and its SYSOP each receive a callsign, and the SYSOP is signed in.

CAUTION	The account created at first-run setup is the station's only SYSOP account. Record its password safely. StationNet provides no means of creating further SYSOP accounts through the station's pages.

34. Starting, Stopping, and the Station Manager

34.1 Running a Single Station

`run.bat`, `./run.sh`, or a packaged `start.bat`/`start.sh` starts the `default` station in a console window. Press Ctrl+C in that window to stop it. To run another station from the command line, give its identifier:

```
python run.py --station <id>
```

A station that is already running cannot be started a second time.

34.2 The Station Manager

The Station Manager (`manager.bat` or `manager.sh`) is a desktop window that lists every station under `stations/` with its name, status, port, and addresses. Each station card provides:

`Start`, `Stop`, `Restart`	Control the station. Stations started here continue running if the manager window is closed. A station that has not stopped within 20 seconds offers `Force Stop`.
`Open`	Opens the station in the browser.

Logs	Opens the station's log.
Settings	Changes the name, port, network access, and the **Served through an HTTPS proxy** option. Changes to a running station take effect when it restarts; `Save & Restart` does both.
Hours	Sets opening and closing times (Chapter 36).
Delete	Removes the station permanently, after you type its name to confirm.

To create a station, choose `+ New Station`. The manager suggests a name and a free port and asks who may connect. It can also create the first SYSOP account, so that the browser setup page is not needed.

A station shown as **CRASHED** stopped unexpectedly; its last log line is displayed. Start it again, and examine the log if the problem recurs.

35. Network Access

By default a station listens on every network adapter of the host, on port 8080. Other devices connect to `http://<host-address>:8080`. If they cannot, permit incoming TCP connections on the station's port in the host's firewall.

In the Station Manager's **Settings** you may restrict a station to **this computer only**, to all adapters, or to one chosen address. For source installations, the environment variables `STATIONNET_HOST` and `STATIONNET_PORT` override the station's settings.

35.1 Secure Connections

Browsers allow the microphone (radio voice), application installation, and PSP emulation only on secure connections. To provide them to other computers, place StationNet behind an HTTPS reverse proxy with a certificate trusted by every client. For example, with Caddy:

```
stationnet.lan {
    tls internal
    reverse_proxy 127.0.0.1:8080
}
```

Set the station to listen on this computer only, select **Served through an HTTPS proxy** in the Station Manager (or set `STATIONNET_HTTPS=1`), make `stationnet.lan` resolve to the host, and install the proxy's root certificate on each client.

> **CAUTION** Enable the HTTPS proxy setting only when clients actually use the `https://` address. With it enabled, sign-in does not work over plain `http://`.

Live voice travels directly between browsers and is designed for clients on the same local network.

36. Station Hours, Sleep, and Wake

Station hours are set from the Station Manager's `Hours` button and take effect at once. They are kept only while a Station Manager window is open (it may be minimized).

- **Closing.** For a few minutes before closing (five by default) the card shows a countdown and a `Stay Online` button, which keeps the station open until its next opening time. Otherwise the station is stopped cleanly at closing time.
- **Opening.** A stopped station is started at its opening time. A station you stop by hand during its hours stays stopped until the next opening.
- **Messages.** The header line, and the *Extended* and *Early Open* messages, may be edited.
- **Sleep.** With SLEEP selected, the computer is put to sleep after the station closes, following a `Cancel Sleep` countdown, unless another station is still running.
- **Wake.** The manager sets a wake timer before the earliest next opening. **Sleep & Wake** in the manager's toolbar sets the countdown and lead time and shows the next scheduled wake.

On Windows, wake timers must be allowed in **Power Options → Sleep → Allow wake timers**, and some computers using Modern Standby ignore them. On Linux and macOS a `wake_command` must be configured in `manager.json`; see the README.

37. SYSOP Control: Station Settings

The `SYSOP CONTROL` tab opens the station settings. A sidebar on the left lists the categories; choose one to show its settings, and choose an entry beneath it to jump to a particular card. On a narrow screen the categories appear as a row of buttons above the settings. **Find a setting**, at the top of the sidebar, searches every card by name, field label, or member name; press Enter to open the first result.

Category	Contents
Overview	Station status at a glance and links to the management consoles
Identity & Appearance	Station name and description, landing page message, station icon, site theme, masthead and ASCII banner, header taglines, theme library, station MIDI theme
Access & Members	Registration and guest access, invitations, permission groups, group membership
Modules	The module switchboard
Community	Events, User Feeds, Imageboard, Office
Media & Content	File archive, Videos, Library, Arcade, WIKI
Radio & Broadcasting	Band allocations, hosted broadcasting, broadcaster storage
Television	TV settings and the link to TV management
MUD	General settings, worlds, races and classes, skills, abilities and effects, dungeon themes, Dungeon of the Day

Saving. Each form has its own save button directly beneath the fields it saves, and nothing else is saved with it. After saving, the result appears at the top of the same card. Blocks marked `APPLIES IMMEDIATELY` (such as generating an invitation, enabling a world, or downloading a core) act as soon as their button is pressed. A form that has been edited but not saved is marked `UNSAVED CHANGES`, and its category shows an amber dot in the sidebar. Moving between categories keeps the edits; saving a different form or leaving the page asks for confirmation first. `CANCEL CHANGES` restores a form's saved values.

Module switchboard. Select the services to offer and choose `SAVE MODULE AVAILABILITY`. A disabled module disappears from navigation, but its data is kept and its settings can still be changed. Each service's settings card shows whether the module is on, with a link back to the switchboard; this is the only place modules are turned on or off.

Identity & Appearance. Here you set the station name and description; the landing page message (Markdown is accepted); the station icon (and whether it is used for the installed application); the site theme, and whether members may choose their own; the masthead theme and the ASCII banner text, lettering style, size, and colour, with a live preview; the header taglines and their colours and effects; and a station MIDI theme tune and its volume. Custom colour themes are imported from JSON files, or removed, in the **Theme library**.

Registration and guest access. Choose the registration mode (Open, Private, or Disabled) and whether guests may browse. For a private station, choose `GENERATE INVITATION`. The complete invitation code and link are shown once only; send them to the new member. Unused invitations may be revoked.

Service settings. The Community, Media & Content, Radio & Broadcasting, Television, and MUD categories configure each optional service. The file archive policy (who may upload, total storage, and largest file) is under Media & Content. These services are described in Chapter 39.

Management consoles. Moderation, TV management, automated broadcasts, Programs, announcements, and World Forge have their own pages. They are linked from the Overview and from the foot of the sidebar.

38. Accounts, Permissions, and Moderation

38.1 Delegating Duties

There is one SYSOP account. Duties are delegated to members through *groups*, which are managed under **SYSOP CONTROL → Access & Members**:

- **Permission groups.** Create a group with `CREATE GROUP`. Give it `HOST BROADCASTS` (radio hosting) or `MODERATE IMAGEBOARD` (pin, lock, and delete on the imageboard), and add radio access rules (`FULL`, `LISTEN`, or `DENIED` for a band, range, or frequency).
- **Group membership.** Open a member and select the groups they belong to. Every member also belongs to the permanent `DEFAULT` group. Across groups, `DENIED` overrides `FULL`, and `FULL` overrides `LISTEN`.
- **Events.** Event creation may be limited to selected groups.
- **MUD building.** In World Forge (**Live world → Builder access**), the SYSOP may authorize members as builders. Builders may edit and publish any authored world; only the SYSOP may change a world's progression or starting room, and new worlds stay hidden until the SYSOP enables them.

38.2 Moderation

The `MODERATION` tab lists accounts, current restrictions, **Open reports**, and a searchable **Audit history**. For each report, read the reported item and choose `RESOLVE` or `DISMISS` with a note.

Open an account to record an action. Actions are formal warnings (delivered by private mail), temporary suspensions, permanent bans, chat mutes, and reinstatement. A reason is required; private notes are visible only to SYSOPs. `DELETE USER CHAT` removes the member's recent public chat messages. The account page also allows an account to be deleted.

A suspended or banned member cannot sign in and is shown the reason. Temporary restrictions expire automatically.

38.3 Passwords

When a member uses `FORGOT PASSWORD`, the station mails a temporary password to the SYSOP. Confirm the member's identity before passing it on.

39. Managing Modules and Content

Bulletins and announcements. Remove any bulletin, reply, or chat message from its ▾ menu. Post announcements with `NEW ANNOUNCEMENT +`.

Videos, Library, and Arcade. Each has a **Community** mode (members upload, within per-member quotas) and a **Curated** mode (station uploads only). Set the default quota and per-member overrides, and upload station material, in the service's settings card. The Videos module also has a maximum video file size setting for member and station uploads, configurable from 1 to 1,048,576 MB. The Library has categories, an option to allow downloads, and a maximum file size. The Arcade section manages emulator *cores*: each platform's preferred core can be chosen, and a missing core can be downloaded while the host has Internet access. PSP uploads have their own size limit.

Television. Set the schedule time zone, upload size limit, and default channel, then choose `OPEN TV MANAGEMENT`. There you create channels (`+ NEW CHANNEL`), build the **Programming library** by uploading videos or by copying them into the import folder and choosing `SCAN IMPORT FOLDER`, and arrange programmes into *blocks* on the **Schedule**. Weekly blocks repeat; one-off blocks override them. The **Auto scheduler** fills a channel's schedule from categorized programmes; review the proposed schedule and choose `PUBLISH`. Nothing goes on the air until published.

Radio. Set band names, ranges, and permitted transmission types under **Band allocations**; assign each broadcast host a frequency and set the maximum continuous session under **Hosted broadcasting**; and set audio storage quotas under **Broadcaster storage**. `MANAGE AUTOMATED BROADCASTS` schedules programmes to start at set times or intervals.

WIKI. Place `.zim` reference files in the station's `kiwix_data` folder (the path is shown in the WIKI settings card) and turn WIKI on in the module switchboard. New files are found on the next visit. Install only collections from sources you trust.

Programs. Choose `PROGRAMS` under the management consoles (**Manage programs**). Upload a program package ZIP and choose `VALIDATE + INSTALL`. Installed programs can be disabled, re-enabled, or uninstalled. Built-in programs can be disabled but not removed. Uninstalling keeps members' saved data.

MUD. Enable the MUD module, then enable individual worlds under **MUD service → Worlds**. Worlds are imported from ZIP files; several may run at once. The section also imports races, classes, skills, abilities, and dungeon themes, and offers an example world to download. World Forge (Section

20.10) builds and edits worlds in the browser.

40. Backup, Upgrade, and Maintenance

40.1 Backing Up

To back up a station, stop it and copy its entire folder under `stations/`. That folder contains the station's identity, accounts, messages, uploads, installed programs, themes, and worlds. Reference collections (`.zim` files) in `kiwix_data` are also kept there and may be very large.

The Station Manager's `Delete` command offers **Back up the station first**, which saves a ZIP of the folder, without `.zim` files. To restore a backup, unzip it into `stations/`.

> **WARNING** A station folder contains every member's private mail and the station's secret key. Store backups securely and never publish them.

40.2 Upgrading

Back up `stations/` first. Stop every station. Then:

- **Windows package:** unzip the new release and run its `install.bat`. The software is replaced; `stations` is left alone.
- **Linux package:** unzip the new release and run its `install.sh`.
- **Source:** replace the source files, keeping `stations/`, and run the installer again.

Installations older than the `stations/` layout are migrated automatically on first start. `python run.py --migrate-only --dry-run` previews the migration.

40.3 Uninstalling

`uninstall.bat` or `./uninstall.sh` offers to back up station data, then removes the generated environment and station data, leaving the StationNet software ready to be installed again.

40.4 Shutting Down Safely

Always stop a station with `Stop` in the Station Manager, or Ctrl+C in its console window, before shutting down the host computer or copying the station's folder. The station then closes its database cleanly. Continuous radio broadcasts and similar activities are restarted automatically at the next start.

PART VII

Developer's Reference

This part introduces the means of extending StationNet. Detailed references are listed in Chapter 45.

41. Project Organization

StationNet is a Python application built on Flask, served by Uvicorn, with data in SQLite. Pages are rendered on the server, with JavaScript where needed. No front-end build step is required to run it; the Office programs and some other components are supplied as prebuilt bundles.

Location	Contents
`run.py`	Starts a station
`stationnet/`	Application package: one module per service
`stationnet/modules.py`	Module switches and navigation names
`stationnet/db.py`	Database schema and migrations
`stationnet/templates/`	Page templates, one folder per service
`stationnet/static/`	Style sheets, scripts, and bundled assets
`stationnet/application_packages/`	Built-in programs and the program API reference
`stationnet/office_*/`	Sources of the Office programs
`tests/`	Automated tests
`packaging/`, `build-all.ps1`	Distribution builders
`android/`	Android host application

Each service is a Flask *blueprint*. A new service is registered in the same manner and should use the common page template (`base.html`) and the module switch mechanism in `modules.py`.

42. Programs and Packages

The most direct way to extend a station is to write a program for the Program Manager. A program is a ZIP file containing `manifest.json` and a JavaScript entry point (or a BASIC `.bas` file). It runs in an isolated worker with no access to the page, the network, or the file system. A minimal manifest:

```
{
  "package_format": 1,
  "slug": "my-application",
  "name": "My Application",
  "description": "A short explanation.",
  "author": "Your name",
  "version": "1.0.0",
  "category": "Utilities",
  "entrypoint": "main.js"
}
```

and a minimal program:

```
StationNet.start((ui) => {
  ui.render(ui.title("Hello"), ui.text("Welcome."));
});
```

The runtime provides interface elements (text, menus, forms, tables, calendars), a canvas with sprites and framebuffers, keyboard, pointer, touch gamepad, and fullscreen support, sound and General MIDI playback, per-user and shared storage, score tables, and multiplayer sessions. Packages are limited to 64 files and 5 MiB compressed. Place `manifest.json` at the root of the ZIP.

The SYSOP installs packages through **Manage programs**. The complete reference is `stationnet/application_packages/APIREADME.md`.

43. Themes

A colour theme is a JSON file defining a **site** palette, a **masthead** palette, or both. Start from the template copied to the station's `themes/` folder (also `stationnet/theme_templates/template-theme.json`), give it a unique ID, name, and author, and edit the colours. The SYSOP imports it under **Station settings → IMPORT COLOUR THEME**. The format is described in `stationnet/theme_templates/THEMES_README.md`. Built-in palettes are defined in `stationnet/themes.py`.

44. MUD Worlds and Content

MUD worlds are distributed as ZIP files containing rooms, exits, items, NPCs, dialogue, quests, triggers, wilderness, dungeons, and their own races, classes, and skills. They may be built in World Forge and exported, or written directly as files. The complete format reference is the creator handbook, available from World Forge's **Handbook** tab. Races and classes, skills, abilities and effects, and dungeon themes may also be distributed separately as JSON. Each kind has a downloadable template in **SYSOP CONTROL → MUD**.

45. Testing, Building, and Further References

Tests. Install the development requirements and run the test suite:

```
python -m pip install -r requirements-dev.txt
python -m pytest
```

Browser-side tests are run with the `npm run test:*` scripts in `package.json`.

Rebuilding bundles. After changing an Office program or the MIDI engine, run `npm ci`, then the matching script: `npm run build:documents`, `build:sheets`, `build:presentations`, `build:draw`, `build:agenda`, `build:basic`, or `build:midi`.

Distribution. `build-all.ps1` builds every package the build computer can produce and places them in `dist/`, where the `GET YOUR OWN STATION` page offers them.

Further references.

Document	Subject
`README.md`	Configuration, services, and networking
`DISTRIBUTION.md`	Building and installing packages
`BROADCASTING.md`	Radio programmes, automation, and interfaces
`stationnet/application_packages/APIREADME.md`	Program packages and runtime
`MUD_CREATOR_HANDBOOK.md`	Complete guide to MUD world creation
`stationnet/mud_world_templates/README_CUSTOM_WORLDS.md`	World file format
`stationnet/theme_templates/THEMES_README.md`	Colour theme format

Appendices

Appendix A. Keyboard Reference

Throughout the station

Key	Action
/	Move to the navigation bar (MUD: the command line)
Esc	Release focus; leave immersive program view
Tab	Move between controls

Chat

Key	Action
Enter	Send message
Shift+Enter	New line

Radio

Key	Action
V (hold)	Talk on a voice frequency
Space (hold)	Morse key, after BEGIN CW TRANSMISSION
Enter	Tune the frequency typed

Television: see Section 14.3.

Library (PDF)

Key	Action
← →	Previous and next page

Program Manager

Key	Action
↑ ↓	Select program
Enter	Open program

MUD

Key	Action
Enter	Send speech or command
Tab	Accept suggestion
↑ ↓	Recall earlier commands
Esc	Close suggestions

Presentations (while presenting)

Key	Action
→ , Page Down, Space	Next slide
←	Previous slide
Esc	End presentation

Appendix B. Quick Reference

Task	Procedure
Sign in	LOGIN; enter username and password
Sign out	STATION; [DISCONNECT]
Change password	[PROFILE]; [EDIT PROFILE]; Change password
Change colours	[EDIT PROFILE]; Theme preferences
Send mail	MAIL; COMPOSE +
Chat privately	USERS; DIRECT CHAT (member must be online)
Post a bulletin	BULLETIN BOARD; NEW BULLETIN +
Add a poll	NEW BULLETIN +; ADVANCED; Add a poll
Share a file	FILES; UPLOAD FILE +
RSVP to an event	EVENTS; choose event; GOING or INTERESTED
Report content	Open the item's ▾ menu, choose `REPORT`; give a reason
Listen to radio	POWER: OFF; SCAN >
Watch television	TV; POWER; GUIDE
Bookmark a book	While reading, BOOKMARK
Search the WIKI	WIKI; select collection; SEARCH
Pin a program	PROGRAMS; choose program; PIN TO FAVORITES
Start in the MUD	MUD; CREATE CHARACTER; /help
Unlock Office	Enter OFFICE Password
Export a document	Documents; EXPORT; choose format
Run BASIC	BASIC; RUN ▶

Appendix C. Troubleshooting

The station address does not open. The station may be stopped or outside its hours, the address or port may be wrong, or the host's firewall may be blocking connections. Confirm the address with the SYSOP. SYSOPs: check the station's status in the Station Manager and permit the port in the firewall.

"Invalid username or password." Usernames and passwords are case-sensitive. Use `FORGOT PASSWORD` as described in Section 2.5.

"Account suspended" or "Account banned." A SYSOP has restricted the account. The message shows the reason and any expiry time.

A service described in this manual is missing. The station does not offer it, or it requires you to sign in.

The radio or a video is silent. Click anywhere on the page; browsers block sound until you do. Check `MUTE`, the volume, and your device's volume.

Live voice reports that the microphone is unavailable. Microphones require a secure connection. Use the station's `https://` address, if it has one, or use the host computer itself. Allow microphone access when the browser asks.

"Your chat is muted." A SYSOP has muted your chat. You may still read.

A direct chat has disappeared. Direct chats are erased when either member disconnects. This is normal.

Office will not unlock. The OFFICE Password is different from your StationNet password. If it has been lost, saved Office files cannot be recovered.

The browser offers no "Install" command. Installation requires a secure address. See Chapter 30.

A PSP game will not start. PSP games require a secure connection or the host computer, and WebGL 2.

WIKI reports LIBRARY UNAVAILABLE. SYSOPs: place `.zim` files in the folder shown in the WIKI settings section.

The Station Manager shows CRASHED. Read the last log line shown, open `Logs`, and correct the cause, such as a port already in use. Then start the station again.

A station will not start: port in use. Another station or program is using the port. Choose a different port in `Settings`.

Appendix D. Modules and Permissions

The following table lists each module, its condition when a station is first set up, and who may use it.

Module (tab)	Initially	Guests	Members	Reserved to SYSOP or delegates
BULLETIN BOARD	On	Read, if permitted	Read, post, reply	Remove any post
ANNOUNCEMENTS	On	Read	Read, vote	Post and edit
CHAT	On	—	Station and direct chat	Delete messages, mute
MAIL	On	—	Send and receive	—
RADIO	On	—	Listen; transmit by band rules	Host broadcasts (group)
FILES	On	—	Download; upload if permitted	Upload policy
USERS	On	—	Directory, profiles, feeds	Moderation
PROGRAMS	On	—	Run programs	Install and manage
IMAGEBOARD	Off	—	Post and reply	Pin, lock, delete (group)
EVENTS	Off	Read, if permitted	RSVP; create if permitted	Creation policy
WIKI	Off	Read, if permitted	Read	Install collections
VIDEOS	Off	Watch, if permitted	Watch; upload in Community mode	Curated uploads
LIBRARY	Off	Read, if permitted	Read; upload in Community mode	Categories, curated uploads

Module (tab)	Initially	Guests	Members	Reserved to SYSOP or delegates
ARCADE	Off	Browse, if permitted	Play; upload in Community mode	Cores, curated uploads
TV	Off	Watch, if permitted	Watch	Channels and schedules
OFFICE	Off	—	Private use	—
MUD	Off	—	Play enabled worlds	Worlds; building (delegated)
USER FEEDS	On	—	Post, comment, react	Remove posts

Appendix E. Glossary

Announcement. An official notice posted by a SYSOP.

Band. A range of radio frequencies set aside for a particular purpose.

Bulletin. A public message on the bulletin board, with its replies.

Callsign. A random identifying code assigned to each station and each member.

Collection. One reference work in the WIKI.

Core. An emulator component that runs games for one platform in the Arcade.

CW. Continuous wave; Morse code transmission on the radio.

Direct chat. A private, unrecorded conversation between two online members.

Guest. A visitor who has not signed in.

Host computer. The computer on which a station runs.

Imageboard. A board of picture threads.

Member. A visitor who has signed in to an account.

Module. A component that provides one service and can be turned on or off.

MUD. Multi-user dungeon; a text-based multiplayer game.

NPC. A character in a MUD world controlled by the game.

OFFICE Password. The password that encrypts a member's Office files.

Programme. A scheduled item on television, or a sequence of content on the radio.

PTT. Push-to-talk; holding a control to transmit by voice.

SSTV. Slow-scan television; still-picture transmission on the radio.

Station. One running StationNet service, with its own name, accounts, and data.

Station Manager. The desktop program used to create, start, and stop stations.

SYSOP. System operator; the administrator of a station.

Tagline. A short line of text displayed beneath the station's name.

World. One game within the MUD, with its own characters.

ZIM file. The file format of WIKI reference collections.